Case 46: Wildland-to-urban WU-E transition and isolated receivers

1 Verification purpose

This case verifies the discrete wildland-to-urban (WTU) transition of Wildland–Urban ELMFIRE (WU–E) at a contiguous interface and at an isolated urban receiver. Its primary reference calculation compares accumulated heat with the material fire-thermal property (FTP), requiring a finite exposure time before ignition. The seven-run matrix also characterizes the optional fireline-intensity shortcut and is intended to expose reversed fuel eligibility, excessive reach, topology dependence, a missing heat path, or an incorrect arrival timestamp. The scientific verdict is separate from analysis procedure state: before execution the former is NOT EVALUABLE and the latter is NOT RUN.

The ELMFIRE formulation under examination corresponds to `a2dfbcdf72209733c000e5d3431e44723e15efea`. Five reviewed WU–E components matched that revision exactly. Output timing was reviewed separately from the output procedure at the stated revision, with its SHA-256 identity retained in the case specification. A separately modified copy of that procedure was not asserted to match the revision. These design-review statements do not by themselves establish the identity of a later simulation.

The optional shortcut is NOT EVALUABLE in the compatibility review. The configuration selects interface model 2, but the active transition condition tests whether a separate integer control equals 2. No deterministic assignment of that control was found. The real-valued critical heat-flux threshold is a different quantity and cannot establish the integer control's value. The primary heat/FTP decision does not depend on this selection. The optional shortcut comparisons become PASS or FAIL only after a review matched to the model revision establishes an enabled selection condition and documents where the required integer value 2 is assigned deterministically.

Current result: FAIL. 7 of 7 required simulation conditions have evaluable evidence. Analysis procedure: **COMPLETE.**

2 Mathematical formulation and numerical implementation

The intended WTU path deposits direct flame contact (DFC) and radiation from a burning wildland cell onto an urban receiver, accumulates

$$Q_j(t+\Delta t) = Q_j(t) + [q''_{DFC,j}(t) + q''_{rad,j}(t)]\Delta t,$$

and ignites after the fixed material threshold $F_{TP,crit}$ is met. Thus a sufficiently heated target must have a finite delay $\tau_{WTU} = t_{ign,target} - t_{ign,source} > 0$. The same heat/FTP law then produces urban rate of spread (ROS); no 30,000-kJ branch is part of this reference calculation.

As an optional extension, for source (i_s, j_s) and target (i_t, j_t), the reviewed fireline-intensity transition condition is

$$r = \sqrt{(i_t - i_s)^2 + (j_t - j_s)^2}, \qquad \mathcal{I} = (f_s \neq 91) \wedge (f_t = 91) \wedge (r \leq 1) \wedge (\mathrm{FLIN}_s > 1000).$$

The comparison is strict. A true branch sets the target's WTU flag and skips the ordinary DFC and radiation calculation for that target. In both Runge–Kutta stages, the level-set update forces its $\phi = -1$. The newly burned-cell loop then records TOA $= T$. Initialization assigns every input $\phi \leq 0$ source cell time of arrival (TOA) zero.

The fire behavior fuel model (FBFM) 01 source has five-percent dead one-hour moisture, zero wind, and zero slope. The input-design calculation uses

$$t_r = 384/\sigma, \qquad \mathrm{FLIN} = t_r I_R R(0.3048),$$

then searches adjacent IEEE-754 binary32 adjustment values so the reviewed default-single-precision arithmetic operation order brackets 999, 1000, and 1001 kW m^{-1}. This is only an input reference calculation; acceptance uses the ELMFIRE fireline intensity raster.

The peer-reviewed WU–E publications describe interface heat transfer and heat/FTP spread [1, 2]. The supplied thesis supports this scaling on PDF page 53 (printed page 31), equations (2.42)–(2.43), and gives vegetation heat-release context on PDF page 64 (printed page 42), equations (2.81)–(2.83). The intended equations are documented directly here. Neither published source supplies the special 1000 kW m^{-1} branch, selector linkage, forced ϕ, or dump-name reference calculation; those come exclusively from the pinned source.

3 Derivation of expected behavior

For the primary reference calculation, positive DFC-plus-radiation must be observed at the sufficient-heat adjacent and isolated receivers and ignition must occur after a positive exposure delay. The initial-wildland-source heat-release rate (HRR) suppression is diagnostic, but later propagation within the vegetative patch may create new heat-emitting sources. The complete receiver history, not the initial-source state alone, determines this observation.

For the optional shortcut, the contiguous simulation conditions request source FLIN of 999, 1000, and 1001 kW m^{-1} at an immediately adjacent FBFM91 band. Only 1001 may transition. Isolated targets test 999 and 1001 at one-cell distance and 1001 at two-cell distance; only the one-cell 1001 target may transition. A zero-adjustment target is the spontaneous-ignition control. Thus the exact boolean reference calculation is

$$(0, 0, 1)_{\text{contiguous}}, \qquad (0, 1, 0)_{\text{isolated}}, \qquad 0_{\text{zero}}.$$

Both eligible transitions occur during the first update at $T = 0$: the heat loop sets the WTU flag, both RK stages force the target negative, and the newly-burned loop assigns TOA to the current solver time. Since initial source TOA is also zero, the exact expectation is

$$t_u = 0 \text{ s}, \qquad t_s = 0 \text{ s}, \qquad \tau_{WTU} = t_u - t_s = 0 \text{ s}.$$

Every ineligible target must retain terminal $\phi > 0$ and nodata TOA. The evaluator requires $\phi \leq 0$ and finite TOA to agree; either observable alone cannot create an ignition result.

4 Verification metrics and acceptance criteria

All seven simulation conditions require confirmed successful completion with matching identities for the exact simulation settings, independently calculated expected responses, complete set of prescribed conditions, and review of supported model selections. The ELMFIRE program, simulation record, and all results must also match the recorded identities. All seven conditions must use the same identified model program. Stale or altered evidence is rejected.

The dump ledger is exact, not merely terminal: 61 rows have indices 1 through 61, times 0 through 60 s, and only row 61 is final. This follows the pinned loop, which dumps at current $T = 0, \ldots, 59$ before incrementing time and then makes a final dump after reaching 60 s. Final fireline-intensity and arrival-time fields are identified by the rounded output time, 60 s. The level-set sequence is indexed by output number, so its terminal field is output 61, not output 60. All 61 level-set fields are required. The terminal three rasters must be one-band, aligned 25-by-25 EPSG:32610 grids with the exact missing-value definition and finite non-nodata values. The simulation record must not report a propagation stall.

Metric	Observable	Acceptance rule
Evidence completeness	confirmed model and input identities, 61 output times, and terminal fields	all exact for 7/7 simulation conditions
Source realization	source-mask FLIN and TOA	max FLIN error ≤ 0.5 kW m^{-1}; every source TOA zero
State consistency	designated-target terminal PHI and TOA	$\phi \leq 0$ iff TOA finite
Intended WTU heat	target DFC+radiation histories	positive for sufficient-heat receivers
Intended ignition delay	target TOA minus source TOA	finite and greater than zero
Contiguous shortcut matrix	below/equal/above target state	false, false, true when selector is valid
Isolated matrix	one-cell insufficient/sufficient and two-cell sufficient	false, true, false
First update	eligible target TOA and target–source delay	both exactly zero within 10^{-8} s
Zero control	zero-adjustment target PHI and TOA	$\phi > 0$, TOA nodata

The two intended heat/FTP metrics are verdict-bearing whenever complete run evidence exists. Shortcut-attribution metrics remain NOT EVALUABLE unless the selector preflight passes; they do not control the primary case decision.

5 Simulation configuration and predicted outcome

Parameter	Value	Rationale
Grid	25×25, 10 m, EPSG:32610	at least 10 cells per dimension; edge-buffered targets
Fuel topology	FBFM01 source, FBFM91 targets, FBFM93 background	explicit vegetative/urban eligibility and isolation
Source requests	0, 999, 1000, 1001 kW m^{-1}	zero control and strict threshold bracket
Receiver distance	1 or 2 cells	equality boundary and reach control
Time	start 0, fixed step/max 1, stop 60 s	exact first-update and terminal ledger reference calculation

Output	every-step PHI; final FLIN and TOA	direct state, intensity, and time observables
Dormant ignition	one point at 61 s	keeps the prior-ignition state false through the run, preventing the vacuous zero-ignition stall test without changing 0–60 s physics
Other mechanisms	spotting, crown, randomness off	removes alternate ignition mechanisms

Figure 1 is generated directly from current inputs and is not a simulation result.

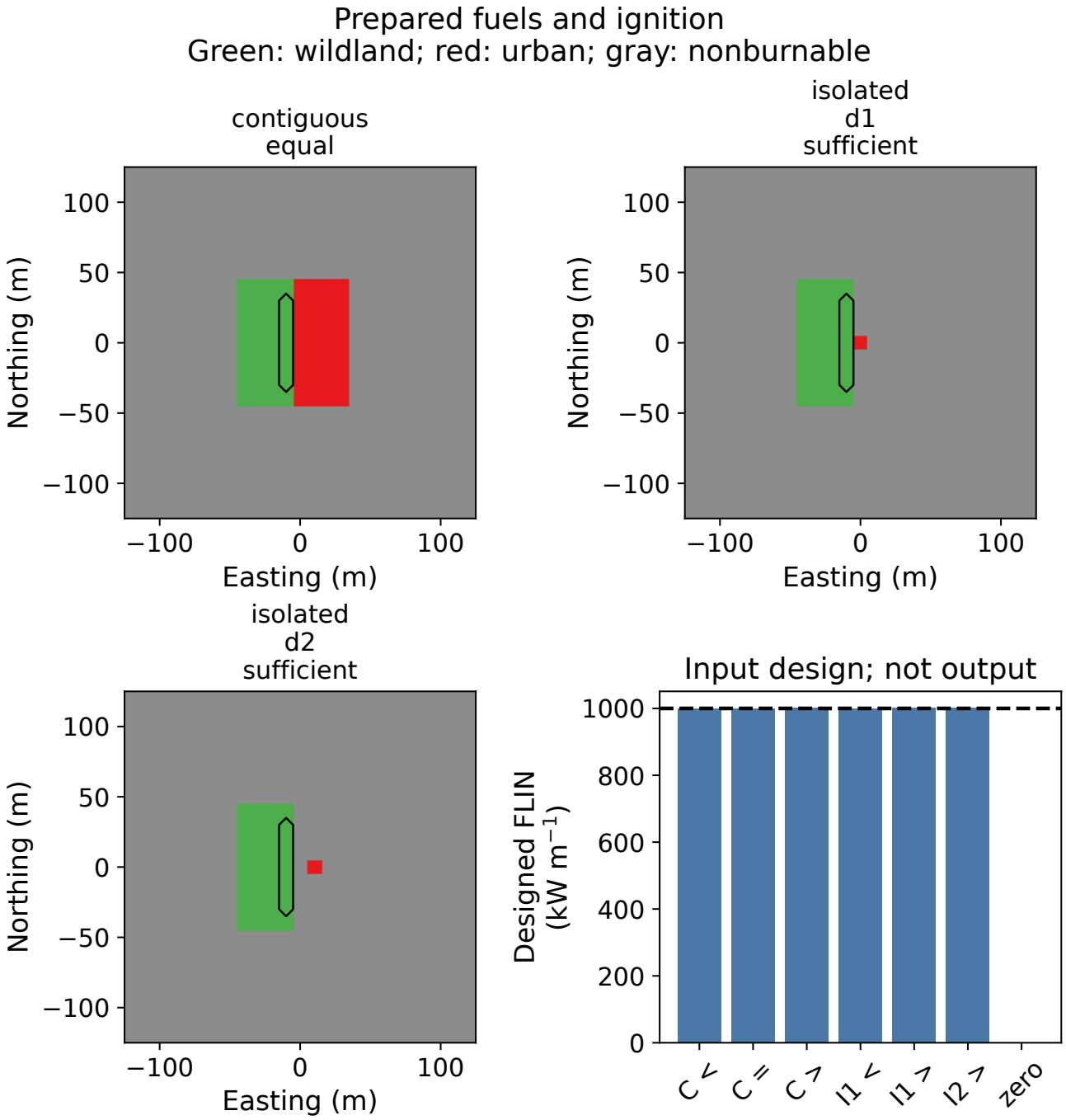


Figure 1: Prepared contiguous and isolated fuel topologies and binary32 source-FLIN design. Green is FBFM01, red is FBFM91, gray is FBFM93, and the black contour marks the seven-cell initial source.

6 Actual simulation results

Analysis procedure status is **COMPLETE**; 7 of 7 simulation conditions have complete current evidence. Selector status is **NOT EVALUABLE**. The generated reason is:

> One or more intended heat/FTP transition checks failed; optional FLIN-shortcut diagnostics do not control this decision.

The executed version is `ELMFIRE 1.1`. The model-program and reference-calculation identities are retained with the calculated metrics.

The primary comparison uses independently calculated expected responses, input-matched simulation results, and the review of supported model settings. Each simulation must have evidence of successful completion. The terminal fireline-intensity and arrival-time fields correspond to 60 s, while the terminal level-set field is the 61st output. Their identities and output times are retained with the calculated metrics so the selection remains traceable.

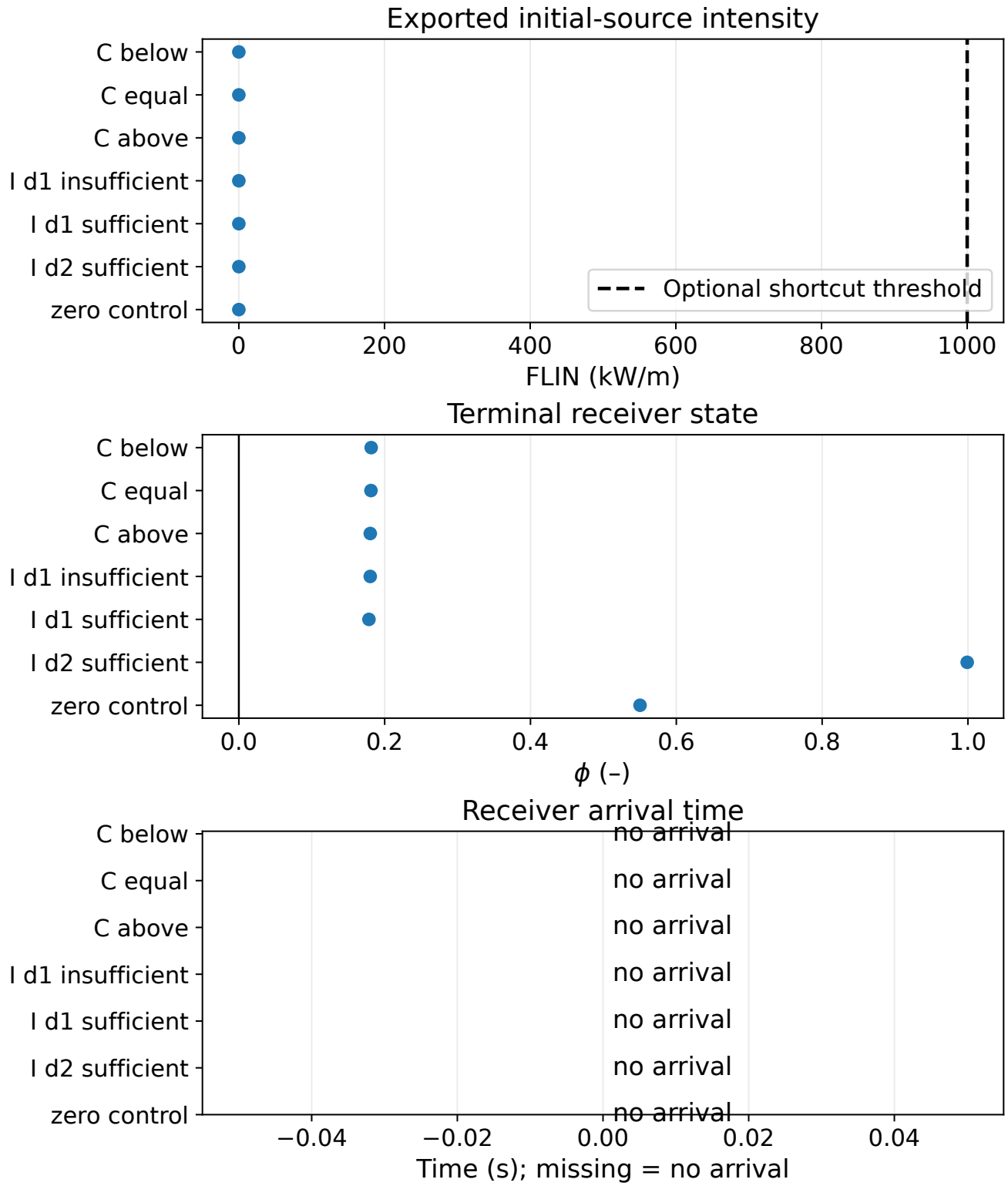


Figure 2: Current black-box source FLIN, terminal target PHI, and target TOA. C denotes contiguous targets and I isolated targets. Missing TOA means no arrival. Only shortcut attribution depends on the selector preflight; the primary checks remain evaluable.

7 Calculated metrics and verification decision

Evidence and exact dump ledger — PASS

Criterion	61 rows: indices 1..61, times 0..60 s, only row 61 final
Observed	7/7 exact; FLIN/TOA time stamp 0000060; PHI dump index d0000061

Source fireline-intensity setup — FAIL

Criterion	maximum absolute error $\leq$0.5 kW/m and source TOA=0 s
Observed	*Units: kW/m.* max abs error kw m: 1001 all source toa zero: Yes

Terminal PHI/TOA state consistency — PASS

Criterion	PHI$\leq$0 iff TOA is finite at each designated target
Observed	contiguous below: Yes contiguous equal: Yes contiguous above: Yes isolated d1 insufficient: Yes isolated d1 sufficient: Yes isolated d2 sufficient: Yes zero control: Yes

Intended W-to-U receiver heat exposure — PASS

Criterion	positive DFC+radiation at sufficient-heat receivers
Observed	*Units: kW/m2.* contiguous above: 61.9232 isolated d1 sufficient: 61.9232

Intended W-to-U finite ignition delay — FAIL

Criterion	finite target TOA with delay $>$ 0 s
Observed	*Units: s.* contiguous above: Not available isolated d1 sufficient: Not available

Contiguous strict-threshold matrix — NOT EVALUABLE

Criterion	below=false, equal=false, above=true
Observed	contiguous below: No contiguous equal: No contiguous above: No

Isolated distance/threshold matrix — NOT EVALUABLE

Criterion	d1-insufficient=false, d1-sufficient=true, d2-sufficient=false
Observed	isolated d1 insufficient: No isolated d1 sufficient: No isolated d2 sufficient: No

First-update transition timing — NOT EVALUABLE	
Criterion	eligible target TOA=0 s and target-source delay=0 s
Observed	*Units: s.* contiguous above: target toa s: Not available target source delay s: Not available isolated d1 sufficient: target toa s: Not available target source delay s: Not available
Zero-source control — PASS	
Criterion	target remains PHI>0 with nodata TOA
Observed	target phi: 0.550248 target toa s: Not available

The authoritative scientific status is **FAIL**. A prepared but unexecuted case is NOT EVALUABLE / NOT RUN. Complete evidence under the selector defect can still evaluate the primary heat/FTP capability; only shortcut-specific rows remain NOT EVALUABLE.

8 Technical interpretation

The current complete run fails the source-intensity setup check. The input reference calculation specifies head-fire FLIN, whereas the terminal raster exports surface-plus-canopy intensity from the initially burned-node list. That export is not the same observation as the head-fire intensity assigned to the WUI source list. A reliable threshold comparison requires a source diagnostic that exposes the intensity actually consumed by the transition branch.

The receiver heat and positive-delay checks can still pass. Although the initial wildland cells have suppressed transient HRR, the surrounding patch can spread and create later heat-emitting cells. Positive receiver heat does not therefore contradict initial-source suppression. Nor does a positive arrival delay alone establish the complete intended accumulated-heat/FTP law.

The optional shortcut has a separate selector defect: the exposed namelist value does not assign the integer tested by the branch. Its attribution rows remain NOT EVALUABLE. The primary heat, delay, state, source-setup, and control rows retain their own decisions; they are not all demoted because the optional selector is unavailable.

9 Scope and limitations

This case covers one wildland fuel/moisture state, one cell size, a 60-s window, contiguous and isolated targets, the strict intensity bracket, distance boundary, state/timing update, and zero control. It does not validate the physical choice of 1000 kW m^{-1}, urban-to-wildland spread, community propagation or convergence. A common model-program identity establishes consistency across accepted outputs but does not establish that the program was compiled from `a2dfbcdf72209733c000e5d3431e44723e15efea`; that source revision identifies the reviewed formulation. A model correction requires a newly reviewed, content-bound PASS preflight before this unchanged strict matrix can yield a shortcut-specific PASS or FAIL.

References

[1] D. M. J. Purnomo et al., "Reconstructing modes of destruction in wildland–urban interface fires using a semi-physical level-set model," *Proceedings of the Combustion Institute*, 40 (2024), 105755. doi:10.1016/j.proci.2024.105755.

[2] D. M. J. Purnomo et al., "Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study," *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.